E-WASTE AND ENVIRONMENTAL MANAGEMENT

Case Study Report

Ten Case Studies on Recycling, EPR and Circular Economy

Name	Hritvi Prasad Narvekar
Class	TE CMPN C
Roll No.	24102C0073
Subject	E-waste and Environmental Management
Submitted to	Dr. Aniket Kundu

Overview of the Ten Case Studies

Each case study follows the prescribed format: Executive summary, Background, Strategy/initiative, Impact & results (data table and graph), SWOT analysis, Comparison with India, Recommendations, Conclusion and References. A one-page exam revision sheet and the consolidated references are at the end.

No.	Case study	Type	Key takeaway
1	Japan's recycling success	International	Law + shared responsibility + public discipline
2	Switzerland's e-waste system	International	Pre-paid ARF, free convenient take-back
3	E-Waste Rules 2022	India - policy	106 items, EPR portal, 60-70-80% targets
4	EPR: success or paper compliance?	India - policy	Capacity 2.2 Mt vs 0.54 Mt recycled
5	Bengaluru	India - city	IT hub; ~90% informal
6	Mumbai	India - city	Informal + formal routes coexist
7	Clean Kerala Company	India - state	31,000 women; waste handled 4x in a year
8	Pune's SWaCH	India - city	Worker-owned co-operative; 900,000+ homes
9	Dharavi	India - cluster	~US$1 bn informal recycling economy
10	Dell's circular strategy	Corporate	50% recycled content, 100% packaging by 2030

Note on data: *Figures are taken from official/industry sources listed in the References. Where sources give estimates (marked '~', 'approx.' or 'est.'), values vary between studies.*

Case Study 1: Why Japan Recycles More Than Most Countries: Lessons for India

1. Executive Summary

Japan's recycling success rests on strict laws, clear shared responsibility (consumer, retailer, manufacturer) and strong public discipline in waste segregation. The Home Appliance Recycling Law (2001) and Small Home Appliance Recycling Law (2013) turn old electronics into a resource - best shown by the Tokyo 2020 medals made from recycled devices. India can adapt the model, but needs enforcement and consumer awareness first.

2. Background

- Japan is resource-poor and short of landfill space, so waste is treated as a resource ('3R': Reduce, Reuse, Recycle).
- Umbrella law: Basic Act on Establishing a Sound Material-Cycle Society (2000).
- Japan generates about 2.6 Mt of e-waste (about 20 kg per person) - UN Global E-waste Monitor 2020.

3. Strategy / Initiative

- Home Appliance Recycling Law (2001): air conditioners, TVs, refrigerators/freezers, washing machines/dryers. Consumer pays a recycling fee at disposal; retailers take back; manufacturers must recycle to legal targets.
- Small Home Appliance Recycling Law (2013): phones, cameras, laptops etc. collected via municipal boxes and authorised recyclers.
- Strict household segregation with municipal schedules (some towns have 40+ categories).
- Tokyo 2020 Medal Project (2017-19): public donated old phones and devices; metals recovered for ~5,000 medals.
- Manifest/tracking system and public awareness from school level.

4. Impact & Results

Indicator	Data
Law	Home Appliance Recycling Law, 2001
Items covered (large)	4 categories (AC, TV, fridge, washer/dryer)
Recycling rate vs target	Achieved rates exceed legal minimum for all items (chart)
Tokyo 2020 medals	~79,000 t of devices incl. ~6.2 million phones
Metal recovered	~32 kg gold, ~3,500 kg silver, ~2,200 kg bronze
E-waste generated	~2.6 Mt (~20 kg per capita)

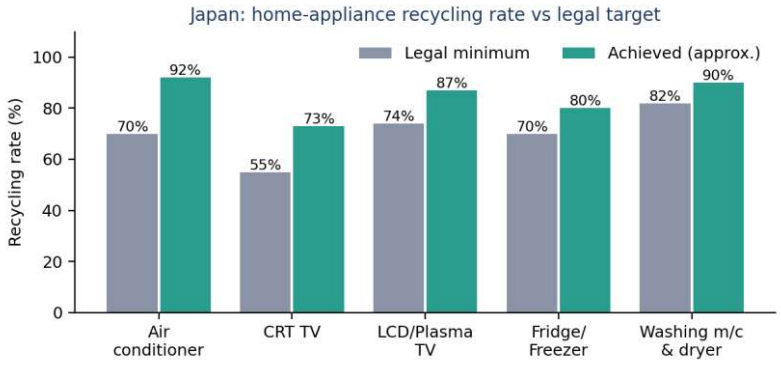


Figure 1: Legal minimum vs achieved recycling rate (approx. values, METI/MOE data).

5. SWOT Analysis

Strengths	Weaknesses
• Clear law with defined roles • High public discipline • Advanced recycling technology	• Consumer pays at disposal - risk of illegal dumping • Municipal recycling only ~20%; much waste incinerated • High operating cost
Opportunities	**Threats**
• Urban mining of metals • Export of technology and know-how	• Rising small-device e-waste • Illegal export of used appliances • Ageing workforce

6. Comparison with India

Aspect	Japan	India
Law	Product-specific, strictly enforced	Broad EPR rules; weak enforcement
Who pays	Consumer fee + producers	Producers via EPR certificates
Segregation	Habit of citizens; detailed	Poor at household level
Informal sector	Negligible	Handles majority of e-waste

7. Recommendations

- Introduce collection through retailers/take-back at point of sale.
- Run school-level awareness on segregation and e-waste.
- Create public campaigns similar to the Medal Project (e.g. 'donate old phones').
- Strengthen enforcement and penalties for illegal dumping.

8. Conclusion

Japan shows that law + responsibility + public habit gives high recycling. India has the rules; what it lacks is enforcement, convenient collection and citizen participation.

Sources: Ministry of the Environment & METI, Japan - Home Appliance Recycling Law annual data. | Tokyo 2020 Organising Committee - Medal Project (olympics.com). | UNU/ITU/ISWA - Global E-waste Monitor 2020.

Case Study 2: Switzerland's E-Waste Success Story

1. Executive Summary

Switzerland runs one of the world's oldest e-waste take-back systems. Consumers return old devices free of charge; the cost is covered by an Advance Recycling Fee (ARF) paid at purchase. Industry bodies SENS and Swico run the system under the 1998 ordinance, achieving about 15 kg of e-waste collected per person per year - among the highest in Europe.

2. Background

- Small, wealthy country with high electronics consumption.
- Voluntary schemes began in the early 1990s (SENS - appliances; Swico - IT/consumer electronics).
- Ordinance on Return, Taking Back and Disposal of Electrical and Electronic Equipment (ORDEE/VREG) made take-back legally mandatory from 1998.

3. Strategy / Initiative

- Advance Recycling Fee (ARF) added to the price of every new product - funds collection, transport and recycling.
- Free return to retailers, manufacturers and public collection points (dense network).
- Industry-run take-back systems (SENS, Swico) with 1,300+ participating manufacturers/importers.
- Certified recyclers audited against technical standards.
- Federal permit needed to export defective appliances; annual public reporting.

4. Impact & Results

Indicator	Data
Collected 2015	133,789 t
Collected 2016	137,808 t
Collected 2017	129,218 t (~15 kg per person)
SENS system, 2021	89,514 t (10.34 kg per person)
Collection rate (some categories)	Up to ~84%
Funding model	ARF - paid in advance by buyer

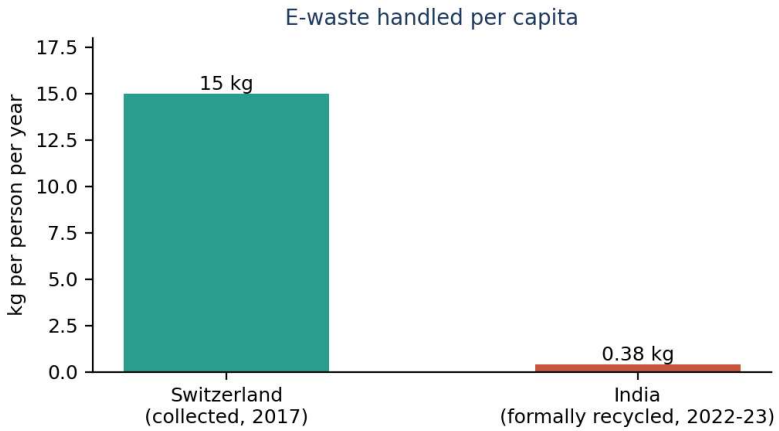


Figure 2: Per-capita e-waste handled - Switzerland vs India (India figure derived: 539,256 t / ~1.42 billion people).

5. SWOT Analysis

Strengths	Weaknesses
• Pre-funded system (ARF) • Convenient free return • Industry ownership and transparency	• Needs very dense network (costly) • ARF must be reviewed to match real costs

Opportunities	Threats
• New streams: solar panels, e-bikes, batteries • Export of the model to other countries	• Li-ion battery fires • Rising volumes • Illegal export of used devices

6. Comparison with India

Aspect	Switzerland	India
Financing	ARF paid at purchase	EPR certificates; no visible fee
Consumer return	Free, convenient	Limited collection points
Collected per person	~15 kg	~0.4 kg (formal)
Operators	Industry non-profits	Producers, recyclers, portal

7. Recommendations

- Pilot an advance recycling contribution on high-volume products.
- Build retailer take-back points in every city.
- Publish annual transparent collection and recycling data.
- Link payments to verified collection, not just recycling.

8. Conclusion

Switzerland proves that pre-paid funding, free convenient return and industry ownership give very high collection. India can adopt these principles in its own EPR system.

Sources: SENS eRecycling - Annual Report 2021 / Facts & Figures (erecycling.ch). | Swico Recycling - Annual Report 2017. | Swiss Federal Office for the Environment (FOEN) - ORDEE ordinance.

Case Study 3: India's E-Waste Management Rules 2022

1. Executive Summary

The E-Waste (Management) Rules 2022 were notified in November 2022 and came into force on 1 April 2023, replacing the 2016 Rules. They widen coverage to 106 products, introduce a digital EPR portal with tradable EPR certificates, and set rising recycling targets (60%, 70%, 80%).

2. Background

- Evolution: E-Waste Rules 2011 -> 2016 -> 2022.
- India generated about 1.6 Mt of e-waste in 2022-23 (CPCB, 21 items under 2016 Rules).
- India is among the top 3 e-waste generators globally; most waste handled informally.

3. Strategy / Initiative

- Scope widened from 21 to 106 electrical and electronic products.
- Mandatory online registration of producers, manufacturers, refurbishers and recyclers on CPCB portal.
- EPR certificates generated by recyclers and traded to meet producers' targets.
- Targets: 60% (2023-24), 70% (2024-25), 80% (2025-26 onwards).
- Refurbishing certificates can defer a producer's EPR obligation (Rule 14(2)).
- Environmental compensation for non-compliance; restriction of hazardous substances (RoHS).

4. Impact & Results

Indicator	Data
Notified / in force	Nov 2022 / 1 April 2023
Products covered	106 (earlier 21)
EPR targets	60% -> 70% -> 80%
E-waste generated 2022-23	1,609,117 t
E-waste recycled 2022-23	539,256 t (~34%)
Authorised recyclers	322; capacity ~2.21 Mt (CPCB)



Figure 3: EPR targets under E-Waste (Management) Rules 2022.

5. SWOT Analysis

Strengths	Weaknesses
• Wider scope • Digital traceability • Market-based certificate system	• Relies on recycler data; limited ground verification • Informal sector not integrated

Opportunities	Threats
• Boost formal recycling and green jobs • Urban mining of metals	• Fake or paper certificates • Price manipulation of certificates • Continued informal burning

6. Comparison with India

Aspect	Rules 2022	Earlier rules / benchmark
Products	106 items	21 items (2016)
Compliance	Online portal + certificates	Authorisation by SPCBs; PROs
Funding	Producers via certificates	Swiss ARF is pre-paid
Consumers	Limited role	Free take-back in Switzerland

7. Recommendations

- Verify certificates with field audits and traceability (QR / tracking).
- Bring informal collectors into the formal chain.
- Add consumer incentives and awareness.
- Strengthen penalties and publish data.

8. Conclusion

The 2022 Rules are a strong framework, but real success depends on actual collection volumes, verification of certificates and bringing the informal sector in.

Sources: MoEFCC - E-Waste (Management) Rules, 2022 (Gazette notification). | Parliamentary replies on e-waste (Rajya Sabha, 1 Aug 2024) - CPCB data. | CPCB - E-waste EPR portal guidelines.

Case Study 4: India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

1. Executive Summary

EPR makes producers responsible for end-of-life of their products. India has built a digital EPR system and a large recycling capacity (~2.2 Mt), but only ~0.54 Mt of the ~1.6 Mt generated was formally recycled in 2022-23. Verdict: a partial success - strong on paper, weaker on ground-level collection and verification.

2. Background

- EPR introduced in India through E-Waste Rules 2011; strengthened in 2016 (targets) and 2022 (certificates).
- Producers must meet targets by getting EPR certificates from registered recyclers.
- Informal sector still handles a large share of e-waste (estimates ~70-90%).

3. Strategy / Initiative

- Producers register and file annual returns on CPCB portal.
- Recyclers upload processed quantity and generate EPR certificates.
- Certificates are sold/transferred to producers to meet targets.
- CPCB audits and issues directions (e.g. Feb 2024 on certificate generation).
- Environmental compensation for default.

4. Impact & Results

Indicator	Data
E-waste generated (2022-23)	1.609 Mt
Formally recycled (2022-23)	0.539 Mt (~34%)
Authorised recyclers	322
Recycling capacity	~2.21 Mt
Capacity used (approx.)	~24%
Informal share of e-waste	Estimated 70-90%

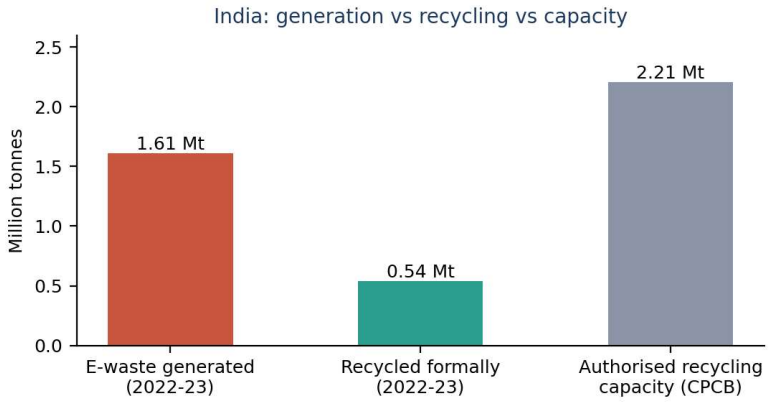


Figure 4: India's e-waste generation vs formal recycling vs capacity.

5. SWOT Analysis

Strengths	Weaknesses
• Digital portal, traceability • Large recycling capacity • Legal obligation on producers	• Capacity underused • Certificates may not reflect true collection • Weak monitoring and informal leakage

Opportunities	Threats
• Integrate informal sector • Blockchain / QR tracking • Third-party audits	• Fake certificates • Recyclers idle, price crash • Hazardous burning continues

6. Comparison with India

Aspect	India	Switzerland / benchmark
What is measured	Recycler output (certificates)	Actual collection per person
Funding	Producer buys certificate	ARF pre-collected
Informal sector	Large, outside system	Almost none
Consumer convenience	Low	High

7. Recommendations

- Measure EPR by verified collection, not only processing.
- Ground-truth audits and penalties for fake certificates.
- Formalise kabadiwalas as authorised collectors.
- Set up convenient retailer take-back.

8. Conclusion

India's EPR is not purely paper compliance - infrastructure exists - but the gap between capacity (2.2 Mt) and recycling (0.54 Mt) shows it needs stronger enforcement and collection.

Sources: Parliamentary Q&A on e-waste (Lok Sabha / Rajya Sabha 2022-24) - CPCB data. | IMPRI - E-Waste Rules 2022: Assessing EPR Framework. | CPCB - EPR portal directions, Feb 2024.

Case Study 5: Bengaluru - India's E-Waste Capital?

1. Executive Summary

Bengaluru, India's IT capital, is among the largest e-waste-generating cities, driven by IT companies and households. It has India's early formal recyclers (e.g. e-Parisaraa) and active NGOs, yet an estimated ~90% of e-waste still reaches informal recyclers.

2. Background

- Home to thousands of IT/electronics firms and a large consuming population.
- Estimates: ~18,000-20,000 t/year (Assocham, older), rising ~20% yearly; other estimates are higher - no reliable city data.
- Karnataka contributes ~9% of India's e-waste (6th among states).

3. Strategy / Initiative

- Early authorisation of formal recyclers by KSPCB (e-Parisaraa, e-Ward, others).
- Karnataka had 71 authorised units with ~52,700 t capacity (CPCB, Lok Sabha reply).
- Bulk consumers (IT firms) channel waste to authorised recyclers.
- NGOs (Hasirudala, Saahas) run collection drives and awareness.
- Informal hub areas such as Nayandahalli.

4. Impact & Results

Indicator	Data
Bengaluru e-waste estimate	~18,000-20,000 t/yr (older)
Karnataka share of India	~9% (6th rank)
Authorised units in Karnataka	71 (capacity ~52,700 t)
Reaches authorised recyclers	~10%
Informal / unaccounted	~90%

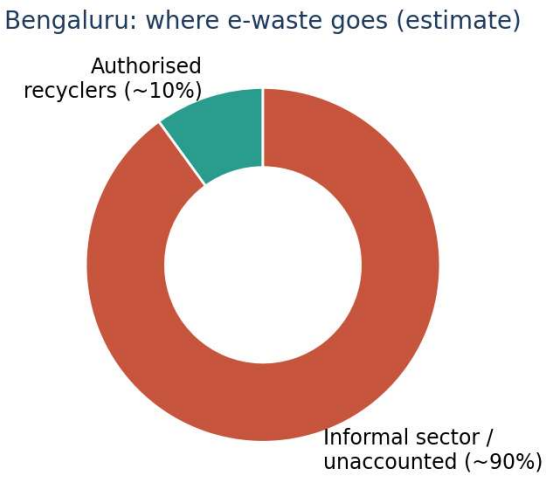


Figure 5: Estimated fate of Bengaluru's e-waste (survey-based estimate).

5. SWOT Analysis

Strengths	Weaknesses
• IT hub and bulk-consumer channel • Pioneer formal recyclers • Active NGOs	• No reliable city-level data • Household waste mostly informal

Opportunities	Threats
• Corporate take-back and EPR • Upgrade informal recyclers	• Burning, acid recovery • Rapid growth (~20%/yr) • Backyard recycling

6. Comparison with India

Aspect	Bengaluru	India
Formal share	~10%	~34% (2022-23)
Main generator	IT bulk consumers	Households + bulk
Recycling units	71 in Karnataka	322 nationwide
Informal sector	Dominant for households	Dominant

7. Recommendations

- Mandatory city-level e-waste inventory.
- Doorstep collection by ward with registered collectors.
- Skill and licence informal recyclers.
- Tie IT companies' CSR/EPR to household drives.

8. Conclusion

Bengaluru is a leading e-waste hub in generation and in formal recycling expertise, but not yet in collection. A city-level data and collection system is the key gap.

Sources: Citizen Matters - Between flames, fumes and bare hands (Bengaluru e-waste). | Deccan Herald - Ctrl+Alt+Del e-waste; India's e-waste problem. | Lok Sabha reply on e-waste (CPCB authorised recyclers, state-wise).

Case Study 6: Mumbai's E-Waste Ecosystem

1. Executive Summary

Mumbai, a financial and port city in India's top e-waste-generating state (Maharashtra), has a layered ecosystem: households and bulk consumers, kabadiwalas, informal dismantling hubs, authorised recyclers, producers under EPR, and regulators (MPCB/BMC). Most e-waste still follows the informal route.

2. Background

- Maharashtra ranks first among states in e-waste generation.
- Mumbai, Delhi and Bengaluru are the top e-waste-generating cities (Assocham ranking).
- Dense population, commerce and electronics markets create large volumes.

3. Strategy / Initiative

- Formal route: producers (EPR) -> collection/take-back -> authorised recyclers (e.g. Ecoreco, Vasai) -> certificates.
- Informal route: kabadiwalas -> dismantling hubs (Govandi, Kurla, Dharavi) -> metals sold.
- Regulation: MPCB authorisation; BMC/NGO collection drives.
- Bulk consumers (corporates, banks) sell to authorised recyclers.

4. Impact & Results

Indicator	Data
Maharashtra rank (states)	1st
Mumbai rank (cities)	Among top 3
Formal route	Producers -> recyclers -> certificate
Informal route	Kabadiwala -> dismantler -> scrap market
Share informal	Most (est. 80-90%)

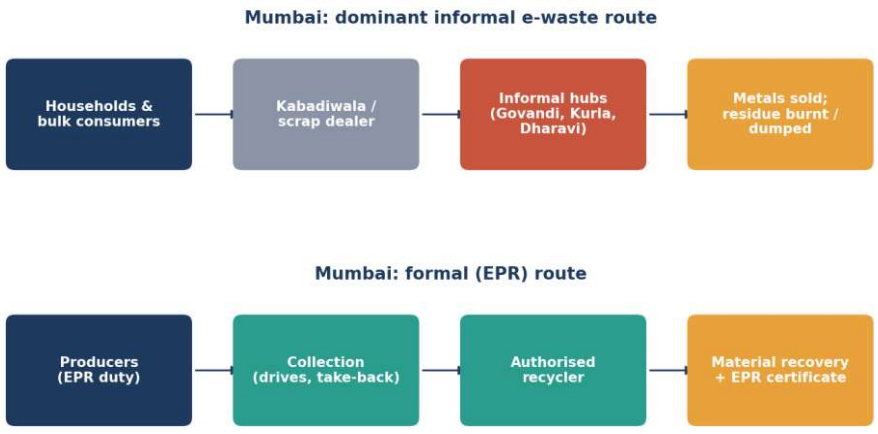


Figure 6: Mumbai's e-waste routes - informal (above) and formal (below).

5. SWOT Analysis

Strengths	Weaknesses
• Large volumes make recycling viable • Presence of established authorised recyclers • Strong scrap-trading network	• Informal dominates • Poor household awareness • Hazardous manual dismantling

Opportunities	Threats
• Link kabadiwalas to EPR • Decentralised collection with BMC	• Illegal imports/dumping • Health risks in slums • Rising lithium batteries

6. Comparison with India

Aspect	Mumbai	India
Generator rank	Top city	3rd largest country
Informal route	Dominant	Dominant
Authorised recyclers	Present (Vasai, Navi Mumbai)	322 nationwide
Policy	Same national rules	Rules 2022

7. Recommendations

- Integrate scrap dealers as registered collectors.
- BMC-run collection bins in every ward.
- Awareness campaigns in housing societies.
- Safety training and PPE for informal workers.

8. Conclusion

Mumbai's ecosystem is large and active, but the informal route dominates. Formal and informal actors must be connected through EPR.

Sources: Assocham - e-waste city ranking (as cited by media). | Deccan Herald - India's e-waste problem: once a trickle now a torrent. | CPCB / MPCB - authorised recyclers list.

Case Study 7: Kerala's Clean Kerala Company

1. Executive Summary

Clean Kerala Company Ltd (CKCL), a state government company, manages non-biodegradable waste collected by Haritha Karma Sena (HKS) - a Kudumbashree women's workforce of ~31,000 members. Waste handled grew from 7,657 t (2021-22) to over 30,000 t (2022-23); HKS collected 66,000+ t in 2024-25. Kerala now also collects e-waste from households at fixed rates.

2. Background

- Kerala has high literacy, high consumption and limited land for landfills.
- Brahmapuram landfill fire (2023) increased pressure for decentralised waste management.
- CKCL created by the Local Self Government Department as a state-owned company.

3. Strategy / Initiative

- HKS members collect dry waste door-to-door (about Rs 50/month households, Rs 100 commercial).
- Waste goes to Mini MCFs / Material Collection Facilities for sorting and baling.
- CKCL buys and transports plastic; shredded plastic used in road tarring and cement kilns.
- Statewide e-waste collection drive (2025): HKS collects household e-waste for a fixed rate.
- Kudumbashree women self-help groups ensure livelihood.

4. Impact & Results

Indicator	Data
HKS members	~31,000
Waste to CKCL 2021-22	7,657 t
Waste to CKCL 2022-23	30,217 t (~4x)
HKS collection 2024-25	66,000+ t
HKS compensation 2022-23	Rs 5.08 crore (2.60 crore in 2021-22)
Kollam district 2025-26	2,797 t; > Rs 29 crore revenue

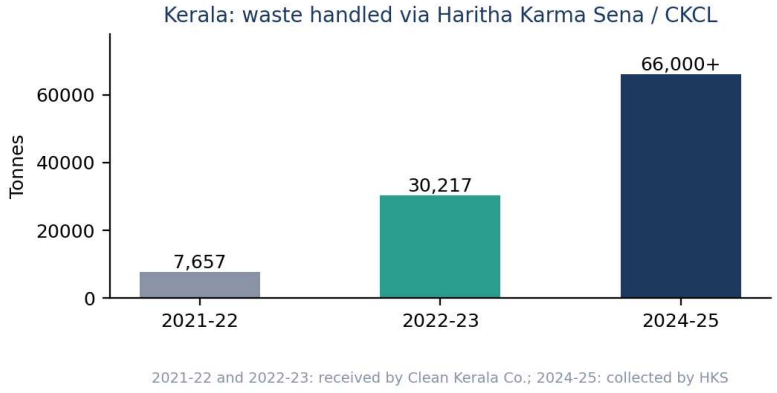


Figure 7: Growth in waste handled through HKS / CKCL.

5. SWOT Analysis

Strengths

- Decentralised, women-led workforce
- Government-backed buyer (CKCL)
- User fee provides income

Weaknesses

- Low margin on non-recyclable plastic
- Dependence on user fee collection

Opportunities	Threats
• Scale e-waste collection • Household incentives for segregation	• Rise in waste volume • Low-value plastic disposal cost

6. Comparison with India

Aspect	Kerala	India
Model	Decentralised HKS + state company	Mostly municipal / informal
Workforce	31,000 women (Kudumbashree)	Unorganised waste pickers
Financing	User fee + CKCL purchase	Varies by city
E-waste	Fixed-rate household collection	No uniform system

7. Recommendations

- Replicate CKCL-type state company in other states.
- Extend HKS to formal e-waste collection with recyclers.
- Digitise tracking of collected waste.
- Link HKS to EPR producers for payments.

8. Conclusion

Kerala shows that a state-backed, women-led, decentralised model can multiply waste collection quickly. It is replicable for e-waste with EPR funding.

Sources: Onmanorama - Tablet strips / Haritha Karma Sena data (2023); Kollam report (2026). | The South First - Kerala e-waste drive (July 2025). | Kerala Government / NGT submission on Kollam waste management.

Case Study 8: Pune's SWaCH Model

1. Executive Summary

SWaCH is India's first wholly worker-owned cooperative of waste pickers, authorised by Pune Municipal Corporation (PMC) for door-to-door waste collection. With ~3,900 members it serves 900,000+ households, handles ~1,000 t/day and makes waste pickers dignified service providers.

2. Background

- Pune waste pickers organised through union KKPKP (1993); SWaCH cooperative formed 2005.
- Before SWaCH, PMC covered only ~7% of households.
- PMC-SWaCH partnership formalised in 2008.

3. Strategy / Initiative

- Members buy shares; work in pairs covering 150-400 households.
- Households pay a monthly user fee (~Rs 50 historically) plus income from recyclables.
- Segregated dry waste sold to recyclers; wet waste composted.
- ID cards, insurance, welfare through KKPKP.
- Services extended to e-waste, clothes reuse, composting.

4. Impact & Results

Indicator	Data
Members	3,500-3,900+ (mostly women)
Households served	~900,000+
Waste handled	~1,000 t/day
Recycled	~70,000 t/yr
Saving to PMC	~US$316,000/yr (transport)
Jobs per tonne	~3.5 (vs 0.1 for waste-to-energy)

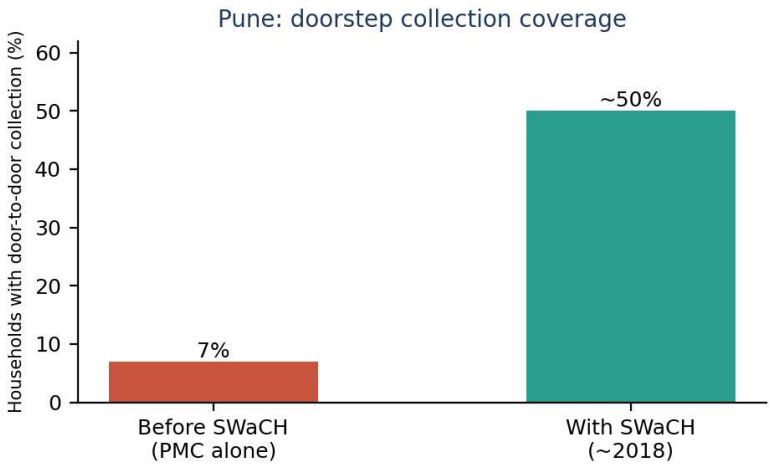


Figure 8: Door-to-door collection coverage in Pune (before vs with SWaCH).

5. SWOT Analysis

Strengths	Weaknesses
• Worker-owned, empowered women • Cost saving for PMC • High recovery of recyclables	• User fee not paid by all households • Dependence on PMC contracts • Unsegregated waste

Opportunities	Threats
• Add e-waste collection under EPR • Scale to other cities	• Waste-to-energy plants displacing pickers • Policy changes

6. Comparison with India

Aspect	Pune (SWaCH)	India
Workers	Formal members, ID cards	Mostly informal, no protection
Coverage	~half of city households	Varies widely
Financing	User fee + recyclable sales	Municipal budgets
Jobs	3.5 per tonne	Waste-to-energy: 0.1

7. Recommendations

- Mandate integration of waste pickers in city SWM plans.
- Link cooperatives to EPR producers for e-waste.
- Make user fee collection easier (property tax link).
- Replicate in other municipalities.

8. Conclusion

SWaCH proves that integrating waste pickers into municipal systems improves collection, saves money and gives dignity. A strong template for e-waste collection.

Sources: WRI - Urban Transformations: Pune's waste pickers. | Resource Recycling (2018) - Picking dignity. | CENFA - SWaCH case study; Down To Earth - What a bright idea.

Case Study 9: Dharavi Recycling Ecosystem

1. Executive Summary

Dharavi in Mumbai is one of Asia's largest informal recycling clusters. About 15,000 small units and hundreds of thousands of workers collect, sort, wash, shred and remelt plastics, metals, paper and also dismantle e-waste, with an estimated turnover of ~US$1 billion a year - but under unsafe conditions.

2. Background

- ~2.1 km2 area with 0.7-1 million residents.
- Recycling is estimated to employ up to ~250,000 workers.
- Redevelopment project (awarded to Adani Group, 2022) raises concerns for existing industries.

3. Strategy / Initiative

- Door-to-door collection by rag-pickers and scrap dealers (kabadiwalas).
- Material-specific units: plastic, aluminium, paper, cloth, leather, e-waste.
- Clustered value chain - each unit does one step; low transport cost.
- Recovered material sold to manufacturers across Mumbai and India.

4. Impact & Results

Indicator	Data
Area / population	~2.1 km2 / 0.7-1 million
Small units	~15,000 single-room units
Recycling workforce	Up to ~250,000 (estimates vary)
Annual turnover	~US$1 billion
Mumbai plastic recycled	~60-80% (estimates)



Figure 9: Dharavi plastic recycling value chain.

5. SWOT Analysis

Strengths • Low-cost, efficient clustered chain • Large livelihood base • High material recovery	**Weaknesses** • Unsafe working conditions • No licences / EPR link • Pollution, health hazards
Opportunities • Formalise and upgrade units • Link to EPR and PPE training	**Threats** • Displacement through redevelopment • Regulatory crackdown without rehabilitation

6. Comparison with India

Aspect	Dharavi	India
Sector	Informal, highly efficient	Informal handles most waste
Safety	Poor	Poor
Legal status	Unlicensed	Authorised recyclers 322
Livelihoods	Large	1.5 million waste pickers (est.)

7. Recommendations

- Integrate units into formal system with licences and safe equipment.
- Preserve livelihoods in redevelopment plans.
- Provide training, PPE and credit.
- Connect to EPR so producers pay for their collection.

8. Conclusion

Dharavi demonstrates a highly efficient, low-cost recycling ecosystem but at high human cost. India's goal should be to upgrade it, not eliminate it.

Sources: Media and academic reports on Dharavi recycling (estimates vary). | Dharavi Redevelopment Project - Maharashtra Government / Adani Realty (2022). | Resource Recycling and urban waste studies.

Case Study 10: Dell's Circular Economy Strategy

1. Executive Summary

Dell aims to keep materials in use through design, recovery and reuse. Its 2030 goals: more than 50% of product content from recycled/renewable material, 100% of packaging from recycled/renewable material, and reusing or recycling an equivalent product for each one sold. Since 2007 Dell has recovered over 2.5 billion pounds of used electronics.

2. Background

- Dell offers recycling services to customers for 20+ years.
- 2020 goals included recovering 2 billion lb of used electronics and using 100 million lb of recycled content - the latter achieved early.
- Circular economy aims to make waste obsolete and cut emissions from new materials.

3. Strategy / Initiative

- Design for circularity: easier to repair, upgrade and disassemble.
- Closed-loop plastics, recovered gold and carbon fiber reuse in new devices.
- Ocean-bound plastic used in products and packaging (443,000+ lb).
- Packaging from recycled paper, bamboo, sugarcane fibers.
- Take-back and recycling programmes to recover materials.

4. Impact & Results

Indicator	Data
Used electronics recovered	2.5 billion lb since 2007
Recycled content goal (2020)	100 million lb - achieved early
Ocean-bound plastic	443,000+ lb recycled
Carbon fiber kept from landfill	2+ million lb
2030 product content goal	> 50% recycled/renewable
2030 packaging goal	100% recycled/renewable

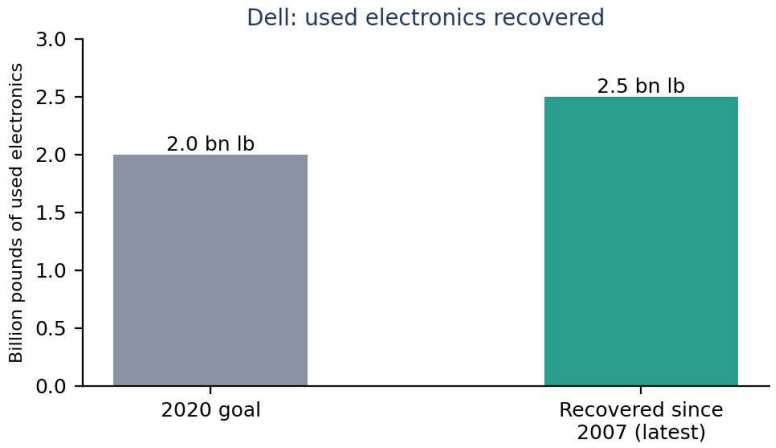


Figure 10: Used electronics recovered by Dell (goal vs recovered since 2007).

5. SWOT Analysis

Strengths	Weaknesses
• Clear public goals • Closed-loop materials • Global take-back network	• Needs product returns to get material • Recycled plastic limited by quality

Opportunities	Threats
• Offer repair and refurbishment • Share closed-loop technology	• Low return rates • Supply-chain complexity

6. Comparison with India

Aspect	Dell	Typical Indian producer
Approach	Design + recovery + reuse	Mostly EPR compliance
Targets	50% / 100% by 2030	Legal 60-80% recycling
Materials	Recycled content in products	Little recycled content
Consumer	Take-back services	Limited take-back

7. Recommendations

- Mandate recycled-content targets for electronics.
- Promote refurbishment and right-to-repair.
- Encourage producers to design for disassembly.
- Incentivise take-back with discounts.

8. Conclusion

Dell shows that producers can move from end-of-pipe recycling to a circular model through design and recovery. Indian producers should shift from certificate compliance to circular design.

Sources: Dell Technologies - 2030 Goals; 'Circular Design Gives E-Waste a Second Life' (blog). | Dell - Sustainable devices and infrastructure brief. | Dell Progress Made Real report.

References

1. Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India - E-Waste (Management) Rules, 2022.
2. Central Pollution Control Board (CPCB) - E-waste EPR portal, guidelines and directions (incl. Feb 2024).
3. Parliament of India - Lok Sabha / Rajya Sabha replies on e-waste (e.g. Rajya Sabha PQ 265, 1 Aug 2024; Lok Sabha AU3395).
4. Forti V. et al. - The Global E-waste Monitor 2020 and 2024 (UNU/ITU/ISWA/UNITAR).
5. SENS eRecycling - Annual Reports 2021 and 2023, Facts & Figures (erecycling.ch).
6. Swico Recycling - Annual Report 2017 (swicorecycling.ch).
7. Ministry of the Environment, Japan and METI - Home Appliance Recycling Law data; Tokyo 2020 Medal Project (olympics.com).
8. IMPRI - E-Waste (Management) Rules (2022): Assessing India's EPR Framework (impriindia.com).
9. Citizen Matters - Between flames, fumes and bare hands: Bengaluru's e-waste (citizenmatters.in).
10. Deccan Herald - India's e-waste problem: once a trickle, now a torrent; Ctrl+Alt+Del e-waste.
11. Onmanorama - Tablet strips plastic waste disposal (2023); Kollam waste recycling (2026).
12. The South First - Kerala launches statewide e-waste collection drive (15 Jul 2025).
13. WRI - Urban Transformations: Pune's waste pickers (wri.org).
14. Resource Recycling - Picking dignity (2018); CENFA - SWaCH case study; Down To Earth - What a bright idea.
15. Dell Technologies - 2030 goals blog; 'Circular Design Gives E-Waste a Second Life'; Sustainable devices brief (dell.com).

One-Page Exam Revision Sheet

#	Case	Key facts / numbers	One-line answer
1	Japan	HARL 2001; small appliances 2013; Tokyo 2020 medals (~79,000 t devices, 32 kg gold)	Law + shared responsibility + citizen discipline
2	Switzerland	SENS & Swico; ARF; ORDEE 1998; ~15 kg/person	Pre-paid fee funds free, convenient take-back
3	Rules 2022	In force 1 Apr 2023; 106 items; EPR portal; 60/70/80%	Wider scope + digital EPR certificates
4	EPR	Generated 1.61 Mt, recycled 0.54 Mt; capacity 2.21 Mt	Partial success; paper-compliance risk
5	Bengaluru	~18-20k t/yr (est.); ~90% informal; 71 units in Karnataka	Generation hub; collection gap
6	Mumbai	Maharashtra #1 state; Govandi/Kurla/Dharavi hubs	Informal route dominates; link to EPR
7	Kerala CKCL	HKS 31,000 women; 7,657 t -> 30,217 t; 66,000 t (2024-25)	Decentralised, state-backed, women-led
8	SWaCH Pune	~3,900 members; 900,000+ households; 1,000 t/day; coverage 7% before	Worker-owned co-op integrated with PMC
9	Dharavi	~15,000 units; ~US$1 bn; up to 250,000 workers	Efficient but unsafe informal cluster
10	Dell	2.5 bn lb recovered; >50% product, 100% packaging by 2030	Design for circularity + take-back

Key terms

EPR = Extended Producer Responsibility. **ARF** = Advance Recycling Fee. **CPCB** = Central Pollution Control Board. **3R** = Reduce, Reuse, Recycle. **Urban mining** = recovering metals from e-waste. **Circular economy** = keep materials in use; design out waste. **Recycling rate** = recycled / generated x 100 (India 2022-23: 0.539 / 1.609 = ~34%).

Quick comparison: India vs best practice

Best practice (Switzerland, Japan): pre-paid funding, convenient free return, strong enforcement, public habit. India: strong rules and capacity, but low collection, large informal sector and limited consumer role. Fix: verified collection, integrate informal workers (SWaCH, Kerala HKS), producer-led circular design (Dell).